Android

1. Applications
2. Application Framework
3. Android Runtime (ART)
4. Hardware Abstraction Layer
5. Linux Kernel

Key Characteristics

- Open Source
- Linux-based
- Java/Kotlin apps
- Highly customizable

iOS

1. Application Layer
2. Cocoa Touch
3. Media Layer
4. Core Services
5. Core OS (Darwin/XNU)

Key Characteristics

- Closed Source
- Unix-based
- Swift/Obj-C apps
- Tightly integrated

Windows

1. Applications & UWP
2. User Mode
3. System Services
4. Windows NT Kernel
5. Hardware Abstraction Layer

Key Characteristics

- Proprietary
- NT Kernel
- C#/.NET/Win32
- Backward compatible

Android

1. **Applications**
   User-facing apps (Java/Kotlin)
2. **Application Framework**
3. **Android Runtime (ART)**
4. **Hardware Abstraction Layer**
5. **Linux Kernel**

Key Characteristics

- Open Source
- Linux-based
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- Highly customizable

iOS

1. **Application Layer**
2. **Cocoa Touch**
   UIKit, Foundation, Event handling
3. **Media Layer**
4. **Core Services**
5. **Core OS (Darwin/XNU)**

Key Characteristics

- Closed Source
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Windows

1. **Applications & UWP**
2. **User Mode**
3. **System Services**
4. **Windows NT Kernel**
5. **Hardware Abstraction Layer**
   Platform-independent hardware interface

Key Characteristics

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Architecture Overview

Android Architecture

Built on Linux kernel, Android uses a layered architecture with the Android Runtime (ART) handling app execution. The HAL provides standardized hardware interfaces.

iOS Architecture

Based on Darwin (Unix), iOS features a hybrid XNU kernel combining Mach microkernel with BSD components. Cocoa Touch provides the primary app framework.

Windows Architecture

Windows NT kernel provides a modular architecture with executive services, supporting multiple subsystems and maintaining extensive backward compatibility.

Operating System Architecture Comparison

A visual comparison of Android, iOS, and Windows system architectures

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